THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 11

OPTIMIZE, Z-Ordering & Liquid Clustering

SECTION 01

The Small File Problem

Tables that receive frequent small writes, such as streaming ingestion or micro-batch jobs, accumulate large numbers of tiny Parquet files over time. This hurts performance because query engines pay per-file overhead (opening files, reading footers, planning tasks) on top of the actual data read.

CORE CONCEPT

More small files means more per-file overhead and slower queries, even if the total data volume hasn't changed. Delta Lake provides three tools to address this: OPTIMIZE for compaction, ZORDER for data clustering within files, and Liquid Clustering as the modern, self-maintaining replacement for both partitioning and Z-Ordering.

SECTION 02

OPTIMIZE — File Compaction

OPTIMIZE compacts small files in a Delta table into larger, more efficient files. It rewrites the data but does not change what the data says, only how it's physically stored.

```
OPTIMIZE sales_transactions;

-- Optionally scope to specific partitions
OPTIMIZE sales_transactions WHERE region = 'APAC';
```

IMPORTANT NUANCE

OPTIMIZE doesn't physically delete the old small Parquet files from storage. It performs a logical remove by marking them as removed in the transaction log, which is why you can still time travel back to versions before the optimize ran. Running VACUUM is what physically deletes those now-unreferenced files.

SECTION 03

Z-Ordering — Data Clustering Within Files

Z-Ordering is a data layout technique, typically run alongside OPTIMIZE, that colocates related data in the same set of files. This dramatically improves the effectiveness of data skipping for queries that filter on the Z-Ordered columns.

```
OPTIMIZE sales_transactions
ZORDER BY (customer_id);
```

CORE CONCEPT

Z-ORDER uses a space-filling curve to map multi-dimensional data into a single dimension while

preserving locality, so that data with similar values across the chosen columns ends up stored physically close together, within the same files.

The Catch With Z-Ordering

Z-Ordering is a point-in-time optimization. As new data is appended, the clustering degrades, because new files aren't automatically Z-Ordered relative to existing ones. Maintaining good clustering requires periodically rerunning OPTIMIZE with ZORDER BY across the entire table, which becomes increasingly expensive as the table grows.

SECTION 04

Liquid Clustering — The Modern Replacement

CORE CONCEPT

Liquid clustering is a data layout optimization technique that replaces both table partitioning and ZORDER, automatically organizing data based on clustering keys. Databricks recommends liquid clustering for all new Delta tables.

```
-- Enable liquid clustering on a new table
CREATE TABLE sales_transactions (
  customer_id STRING,
  region STRING,
  amount DOUBLE,
  txn_date DATE
)
CLUSTER BY (customer_id);

-- Enable it on an existing table
ALTER TABLE sales_transactions
CLUSTER BY (customer_id);
```

Why It's Better Than Z-Ordering

- Incremental maintenance: instead of rewriting the whole table, Liquid Clustering only rewrites the files that have drifted out of an optimal layout, which is far cheaper than a full OPTIMIZE plus ZORDER rerun.
- Flexible keys: clustering keys can be redefined at any time without rewriting existing data, letting the layout evolve as query patterns change.
- No fixed partition columns: it removes the need to choose rigid partition columns upfront, which can otherwise create small-file or data-skew problems of their own.

SECTION 05

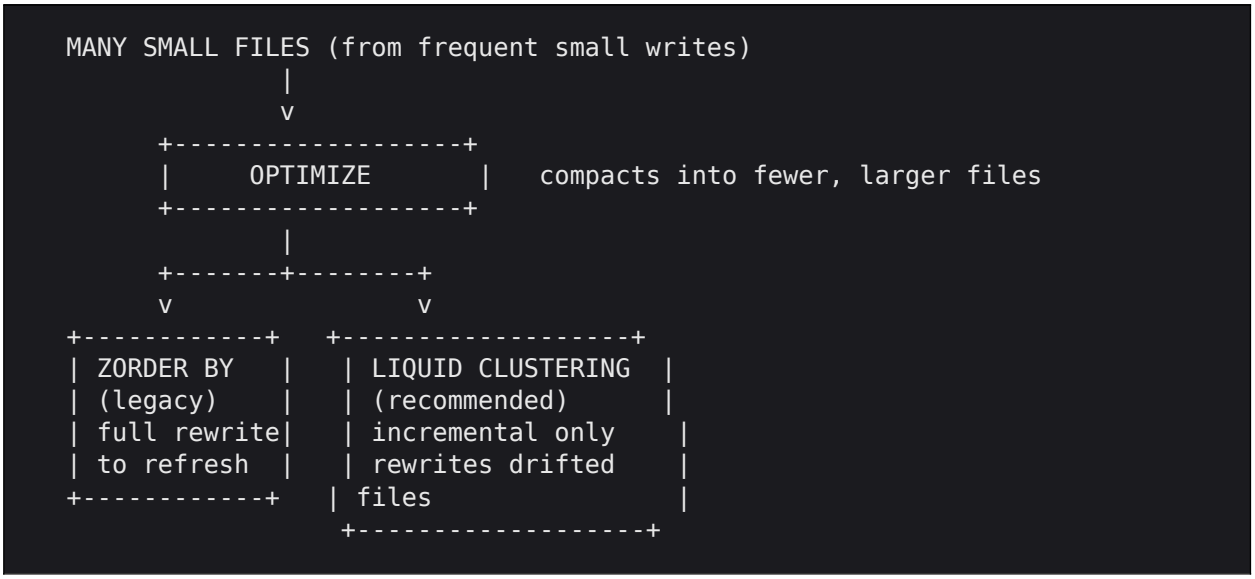
OPTIMIZE vs Z-Ordering vs Liquid Clustering

Aspect	OPTIMIZE (alone)	Z-Ordering	Liquid Clustering

Purpose	Compacts small files	Clusters related data together	Replaces partitioning and Z-Order
Maintenance Cost	Rerun periodically	Full table rewrite to re-cluster	Incremental, only drifted files
Key Flexibility	Not applicable	Fixed at time of run	Can be redefined anytime
Current Recommendation	Still used for compaction	Legacy for new tables	Recommended for all new tables

SECTION 06

How the Three Techniques Relate



REMEMBER THIS FOR INTERVIEWS

OPTIMIZE and ZORDER solve different problems, file count versus data layout, but they're often run together. Liquid Clustering is positioned as the modern successor to both partitioning and Z-Ordering, chosen specifically for its incremental, self-maintaining behavior.

SECTION 07

Glossary — Day 11 Terms

Term	Definition
Small File Problem	Performance degradation caused by too many tiny files, due to per-file read and planning overhead.
OPTIMIZE	The command that compacts small Delta files into larger, more efficient files.

ZORDER BY	A clause used with OPTIMIZE that clusters related data together within files using a space-filling curve.
Data Skipping	The query engine's ability to skip reading files that can't contain data matching a filter, based on file-level statistics.
Liquid Clustering	A modern data layout technique that replaces both partitioning and Z-Ordering, using incremental, flexible clustering keys.
Clustering Key	The column(s) used by Liquid Clustering to organize data, which can be changed at any time without a full rewrite.

SECTION 08

Interview Questions & Answers

Q1. What is the difference between OPTIMIZE and ZORDER?

A. OPTIMIZE compacts small files in a Delta table into fewer, larger files. ZORDER, run alongside OPTIMIZE, clusters related data together within those files so queries can skip irrelevant files more effectively. They solve different problems but are commonly used together.

Q2. Does OPTIMIZE immediately delete the old small files?

A. No. OPTIMIZE performs a logical remove by marking old files as removed in the transaction log, which preserves time travel to earlier versions. VACUUM is what physically deletes those files from storage.

Q3. What is the main limitation of Z-Ordering over time?

A. Z-Ordering degrades as new data is appended, since new files aren't automatically clustered relative to existing ones, requiring an expensive full-table OPTIMIZE with ZORDER BY to restore good clustering.

Q4. What is Liquid Clustering, and why does Databricks recommend it for new tables?

A. Liquid Clustering is a data layout technique that replaces both table partitioning and ZORDER, automatically organizing data by clustering keys. It's recommended because it only rewrites drifted files incrementally, rather than requiring a full-table rewrite, and its clustering keys can be redefined without rewriting existing data.

Q5. Can you change the clustering key on a Liquid Clustering table after data is loaded?

A. Yes, clustering keys can be redefined at any time without rewriting existing data, allowing the data layout to evolve as query patterns change, unlike fixed partition columns.

Q6. Why might a table's queries slow down over time even though the underlying data volume hasn't changed?

A. Because frequent small writes accumulate many tiny files, increasing per-file overhead during query planning and execution. Compacting with OPTIMIZE, and clustering with ZORDER or Liquid Clustering, addresses this.

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